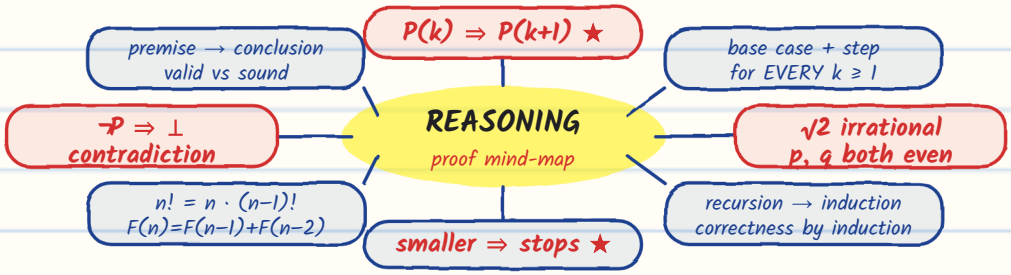
Date: 14.10.2026
topper's notes

Mathematical Reasoning

Induction · Contradiction · Recursion — the three proof machines of the syllabus: exam-ready in 13 sheets



↑ the whole topic on one page: red = high-yield, blue = definitions

➜ Syllabus coverage

"Discrete Mathematics, Theory of Computation and Basic Graph Theory: Set theory; Relations; Functions; Graph theory; Propositional calculus and predicate calculus; Mathematical reasoning: induction, contradiction and recursion; Counting; ..." Sets and functions are in the first lesson, graph theory and the two calculi in the previous one.

High-yield list

① Induction = base case $P(1)$ + step $P(k) \Rightarrow P(k+1)$

② the step must hold for **every** $k \geq 1$, and **forward**

③ $1+2+\ldots+n = n(n+1)/2$; $1+2+2^2+\ldots+2^n = 2^{n+1} - 1$

④ Contradiction: assume **the negation** of P, derive $\perp$

⑤ $\sqrt{2}$ irrational: $p^2 = 2q^2 \Rightarrow$ p, q both even $\Rightarrow$ gcd $\neq 1$

⑥ Contrapositive $\neg Q \rightarrow \neg P$ is **equivalent**; converse is not

⑦ $n! = 1$ $(n = 0)$ else $n \cdot (n-1)!$; $F(n) = F(n-1)+F(n-2)$

⑧ Recursion must **strictly decrease** or it never ends

⑨ naive fib(5) = **15** calls $\approx 1.618^n$; factorial = **6**, $O(n)$ · ⑩ induction $\Rightarrow$ **tree: n ⇒ n − 1 edges**

Legend yellow = key term · red circle = must-memorise fact · **purple box** = memory trick · **red box** = exam trap · **green** = worked example

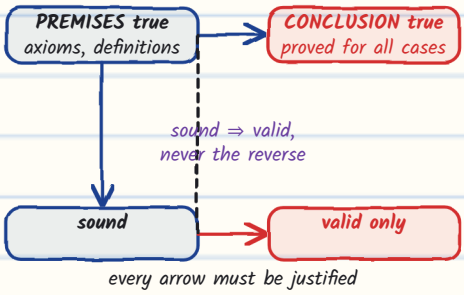
➜ Contents

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✓ Induction: the two parts ... p. 3
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✓ Worked proof: Σ powers of 2 ... p. 5
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1 What a Proof Is

foundation ★

Definition *Mathematical reasoning* is the logical process of deducing conclusions from given statements or **premises**. A *proof* is an argument whose every inference is valid; it is **valid** when the conclusion cannot fail however the premises turn out, and **sound** when the premises were also true. ~~A proof must cover every admissible case, not the ones you tried.~~



sound needs true premises **and** *valid steps; valid needs only the steps*

Method	Shape of the argument	Best for
Direct proof	*P follows from the premises by known rules*	*most theorems*
Contradiction	**assume ¬P**, *derive the impossible*	*"no such thing exists"*
Contrapositive	*P → Q proved as* ¬Q → ¬P — **equivalent**	*implications denying a hypothesis*
Induction	*P(1) holds, and P(k) ⇒ P(k+1) for all k*	*statements about* ℕ
Counterexample	*one failing case kills a universal claim*	*disproving, never proving*

Exam trap A **counterexample** **disproves** *but never proves, and a* **single true instance** *never proves a universal claim. "I checked n = 1 … 10 and it held" is* **not** *a proof about all n — that gap is exactly what induction was invented to close.*

What counts as a premise **definitions**, *which you may unfold inside the proof · axioms of the branch in hand · lemmas you have already proved · the* **hypotheses** *of the statement itself. What does* **not** *count: "it is well known", "the diagram makes it clear", "it checks out for small cases".*

Memory trick — naming the method *Read the* **shape** *of the question, not the topic. "For every n ≥ 1 …" → **induction**. "There is no …" or "show that not …" → **contradiction**. "If … then …" → **contrapositive**. "Is there any …?" → hunt for a **counterexample** to show there is not.*

2 Mathematical Induction: the Two Parts

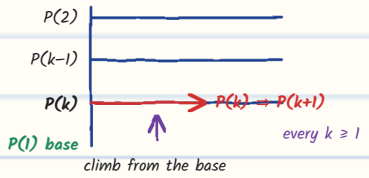
most tested ★

Definition *Mathematical induction* proves a statement $P(n)$ about the natural numbers from a single *principle of mathematical induction*: if $P(1)$ is true **and** $P(k) \Rightarrow P(k+1)$ for **every** integer $k \geq 1$, then $P(n)$ is true for **every** $n \geq 1$. Both halves are compulsory — the *base case* alone proves nothing, and the *inductive step* alone starts from nothing.

Part	What you must write	If you omit it
1. Base case ($n = 0$ or 1)	"For $n = 1$, $P(1)$ is **true**." Check it numerically.	no bottom rung
2. Inductive step	"Let $k \geq 1$ be arbitrary and **assume $P(k)$**; show **$P(k+1)$**."	one case proved
3. Conclusion	"By the principle of mathematical induction, $P(n)$ holds for all $n \geq 1$."	no principle named

Memory trick "**B**ase **B**uilds the first brick; the **S**tep carries the idea one rung further, and it must carry it **every** time." The step is a **rule**, not a **sample** — write "suppose $P(k)$ is true for an **arbitrary** $k \geq 1$", never "let $k = 3$".

Statement $P(n)$ you may be asked to prove	Base case	Shape of the step
$1 + 2 + \dots + n = n(n+1)/2$	$n = 1$	add the term $k+1$ to both sides
$1 + 2 + 2^2 + \dots + 2^n = 2^{n+1} - 1$	$n = 0$	double the whole left-hand side
$n^2 \geq n$ for every $n \geq 1$	$n = 1$	$(k+1)^2 = k^2 + 2k + 1 \geq k^2 + k$
$n(n+1)$ is even for every $n \geq 1$	$n = 1$	$(k+1)(k+2) = k(k+1) + 2(k+1)$
$n \mid n!$ for every $n \geq 1$	$n = 1$	$(k+1)! = (k+1) \cdot k!$



each rung is proved **using the rung below it** *— for an arbitrary k, not a chosen one*

Reading the base case off the statement The first value of n is fixed by the formula, not by taste. $1 + 2 + \dots + n$ starts at $n = 1$; $1 + 2 + 2^2 + \dots + 2^n$ starts at $n = 0$; $n!$ uses $n = 0$. If the claim says "for all $n \geq 0$" the base case **must** be $n = 0$ — starting at $n = 1$ leaves $n = 0$ unproved.

3 Worked Proof: 1 + 2 + 3 + … + n = n(n+1)/2

the standard derivation ★

Worked example ✍ full proof **Let P(n):** 1 + 2 + 3 + … + n = n(n+1)/2. **(1) Base case (n = 1).** LHS = 1, RHS = 1(1+1)/2 = 1, so P(1) is true **(2) Inductive hypothesis.** Let **k ≥ 1** be arbitrary and **suppose P(k):** 1 + 2 + … + k = k(k+1)/2. **(3) Inductive step.** Show P(k+1): 1 + 2 + … + k + (k+1) = k(k+1)/2 + (k+1) (by the hypothesis) = (k+1)[k/2 + 1] = (k+1)(k + 2)/2 = (k+1)((k+1) + 1)/2 — exactly the right-hand side of P(k+1), so P(k+1) is true **(4) Conclusion.** P(1) holds and P(k) ⇒ P(k+1) for every k ≥ 1, so by the principle of mathematical induction P(n) is true for every n ≥ 1. ■

Exam trap — the single highest-yield mistake in this lesson The inductive step must show **P(k) ⇒ P(k+1) for EVERY integer k ≥ 1**. Three failures score nothing. (i) Verifying the step at **one particular value** of k, say k = 3 — that is a sample, not a proof. (ii) Writing the **reverse** implication P(k+1) ⇒ P(k), a completely different statement. (iii) Dropping the words "**assume P(k)**" and simply asserting the result for k+1. If a proof can be summarised as "it held for k = 1, 2, 3, 4, 5", it is not induction.

How to write the step so the examiner is satisfied

(i) name the statement "P(n): …"; (ii) base case, numerically; (iii) "assume P(k) for an arbitrary k ≥ 1"; (iv) compute the k+1 case **starting from the hypothesis** and mark where it was used; (v) finish with "by the principle of mathematical induction". Marks are awarded per line — do not skip (v).

Worked example ✍ why the hypothesis matters

The step for this proof is a one-line substitution, but only **because** "1 + 2 + … + k" was replaced by its closed form k(k+1)/2. Write the line without that substitution and you have merely restated the claim for k+1 — which is ~~exactly what induction asks you not to do~~. The hypothesis is the engine; a step that never calls the engine proves nothing.

Worked example ✍ two lines that look like proofs and are not

(i) "It holds for n = 1, 2, 3, so it holds for all n." This tests **one** value of k in the step — a sample, not a proof. (ii) "P(k+1) ⇒ P(k), hence by induction." This runs the ladder **backwards**: even if the reverse implication is true, it says that knowing a statement at k+1 tells you about k, which never gets you from the base case to the top. Only P(k) ⇒ P(k+1) climbs.

When induction beats direct algebra A claim that carries an **n** on both sides — n!, Fibonacci, running sums, divisibility — cannot be rearranged usefully, and the recursive structure of the statement matches the shape of the step exactly. For a claim like "$n^2 + n$ is always even", a one-line factorisation n(n+1) is shorter and just as correct; the examiner accepts either. Reach for induction when the statement is **built from the previous case**.

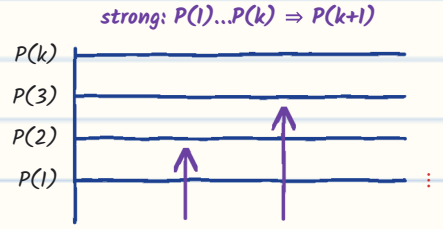
4 Second Worked Proof: $1 + 2 + 2^2 + \dots + 2^n = 2^{n+1} - 1$

Frequently tested ★

Worked example ✍ the standard written answer **Let P(n):** $1 + 2 + 2^2 + \dots + 2^n = 2^{n+1} - 1$, for every integer $n \geq 0$. **(1) Base case (n = 0).** LHS $= 2^0 = 1$, RHS $= 2^1 - 1 = 1$, so **P(0) is true** **(2) Inductive hypothesis.** Let $k \geq 0$ be arbitrary and assume P(k): $1 + 2 + 2^2 + \dots + 2^k = 2^{k+1} - 1$. **(3) Inductive step.** We show P(k+1): $1 + 2 + \dots + 2^k + 2^{k+1} = (2^{k+1} - 1) + 2^{k+1}$ (by the hypothesis) $= 2 \cdot 2^{k+1} - 1 = 2^{k+2} - 1 = 2^{(k+1)+1} - 1$ — this is exactly P(k+1), so **P(k+1) is true** **(4) Conclusion.** P(0) holds and P(k) ⇒ P(k+1) for every $k \geq 0$, so by the principle of mathematical induction P(n) is true for all integers $n \geq 0$. ■

Definition — strong (complete) induction To prove P(n) for all $n \geq 1$ it is enough to show: **(a)** P(1) is true, and **(b)** for every $k \geq 1$, if **P(1), P(2), …, P(k) are all true, then P(k+1) is true.**

Strong induction is ordinary induction with a stronger hypothesis, and the two are equivalent — nothing can be proved by one that the other cannot. It is used exactly when the k+1 case genuinely needs several earlier cases at once.



the k+1 case may lean on all of them

strong induction: several rungs of the ladder may be used at once

Worked example ✍ strong induction **Prove every integer $n \geq 2$ has a prime divisor.** Base n = 2: 2 is prime, so 2 | 2. Step: suppose every integer in 2 … k has a prime divisor and consider k+1. If k+1 is prime the claim is immediate; if it is composite it has a divisor d with $1 < d < k+1$, and by the strong hypothesis d has a prime divisor p, which then also divides k+1. **Hence every $n \geq 2$ has a prime divisor.**

Exam trap (i) Induction proves statements about $\mathbb{N}$ only. (ii) The step must read "**for every $k \geq 1$**"; a candidate who writes "for some k" has not proved it. (iii) The hypothesis must actually be **used** — if the k+1 line follows without it, the argument was circular. (iv) If the k+1 case needs two earlier cases, either use **strong** induction or the statement is false as written.

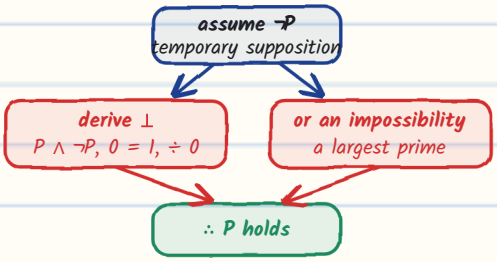
Worked example ✍ a third routine proof **Prove $1 + 1/2 + 1/2^2 + \dots + 1/2^n = 2 - 1/2^n$ for $n \geq 0$.** Base n = 0: LHS = 1, RHS = 2 − 1 = 1 ✓. Step: assume the sum to k is $2 - 1/2^k$; then the sum to k+1 is $(2 - 1/2^k) + 1/2^{k+1} = 2 - 2/2^{k+1} + 1/2^{k+1} = 2 - 1/2^{k+1}$ ✓ — exactly P(k+1). Same three lines, three different statements.

When induction must NOT be used Induction proves P(n) for **natural** n and nothing else. A claim quantified over the reals ("the square of an irrational is irrational" — false) cannot be settled by it, and a claim about **infinitely many** real values is not the kind of statement the principle speaks to. Equally, a claim that fails for one natural n is **disproved**, not proved: no induction can rescue a false statement.

5 Proof by Contradiction

most tested ★

Definition Also called **reductio ad absurdum**. To prove P: **assume ¬P** — the assumption is a temporary **supposition**, not a claim — and from it derive something impossible: a statement together with its negation, 0 = 1, a division by zero, an object with contradictory properties. That impossibility is written ⊥. Since ¬P leads to absurdity, **P must be true**.



anything impossible will do — **one** contradiction finishes the proof

What counts as a contradiction ⊥	Classic use
a statement together with its **negation**	"p even **and** odd"
0 = 1, or any false arithmetic identity	√2 rational
division by zero	1 has no multiplicative inverse
an object with impossible properties	"the largest prime", "a largest integer"
a set both **non-empty and empty**	no bijection $\mathbb{N} \to \mathbb{N} \setminus \mathbb{N}$

Memory trick "Astract the positive: assume the negation, push it to the **absurd**, and the original falls out." The tell-tale of the method is the word **"suppose"** / "assume" followed by something you immediately start attacking.

The recipe, in the four lines an examiner expects **(1)** State the claim P explicitly. **(2)** Write "suppose, for contradiction, that ¬P", and keep the negation as a **supposition**. **(3)** Derive a chain of consequences and push it to an impossibility ⊥. **(4)** Close with "the assumption is false, therefore P" — the contradiction is about the **assumption**, never about P.

Exam trap "A proof by contradiction begins by assuming" — the answer is **the negation of the statement to be proved**. Not "the statement is true" (that is circular reasoning), not "the converse", not "the base case P(1)". And the derivation must end in something **impossible**, not merely **unlikely**: "the evidence points the other way" is no contradiction at all.

When contradiction is the wrong tool If a claim fails for **one** value, do not build an absurd assumption — exhibit the **counterexample** and stop. And a claim about **real** numbers cannot be settled by induction: induction is a theorem about $\mathbb{N}$ only.

6 Contradiction in Action

written favourite ★

Worked example ✍ the canonical proof that √2 is irrational

1. **Suppose, for contradiction, that √2 is rational.** Then $\sqrt{2} = p/q$ with integers p, q, $q \neq 0$ and **$\gcd(p, q) = 1$** in lowest terms.
2. Square both sides: $2 = p^2/q^2$, hence **$p^2 = 2q^2$**.
3. p^2 is even, so **p is even** — an odd p has an odd p^2 — write $p = 2k$.
4. Substitute: $(2k)^2 = 2q^2 \Rightarrow 4k^2 = 2q^2 \Rightarrow q^2 = 2k^2$.
5. q^2 is even, so **q is even**.
6. Then 2 divides both p and q, so **$\gcd(p, q) \geq 2$** — contradicting $\gcd(p, q) = 1$. ⊥
7. Therefore the assumption was false and **√2 is irrational**. ■

Worked example ✍ infinite descent

Claim: no sequence of positive integers can descend for ever. **Suppose** one does: $a_1 > a_2 > a_3 > \ldots$, all positive integers. By **well-ordering** $\{a_1, a_2, \ldots\}$ has a least member a_m. But $a_{m+1} < a_m$ is also in the set and smaller — contradicting leastness.

Contradiction ⇒ every set of positive integers has a least member. This is **infinite descent**, the engine behind "√2 is irrational".

Worked example ✍ a family that fails

Claim: $f(n) = n^2 + n + 41$ is prime for every natural n. **Suppose** it is prime for all n. Take **$n = 41$**: $f(41) = 1681 + 41 + 41 = 1763 = 41 \times 43$, which is not prime.

A single counterexample destroys the universal claim — contradiction and counterexample agree here, which is why one bad case is the cheapest refutation a candidate can offer. ($f(n)$ is prime for $n = 0 \ldots 39$, which is what makes it tempting.)

Exam trap In the √2 proof the **lowest-terms** hypothesis $\gcd(p, q) = 1$ is what makes the final contradiction bite; without it the derivation is circular. Also "p^2 is even ⇒ p is even" is a **lemma** that must be stated: what is immediate from $p^2 = 2q^2$ is only that p^2 is even, and the step to p is the **contrapositive** of "p odd ⇒ p^2 odd".

Contradiction in the computer-science syllabus

The same shape proves that no total computable function decides a Turing machine's halting behaviour — assume the decider D exists, ask D about D, and D must be wrong about itself. Expect the written papers to ask you to **name the assumption** and **point at the contradiction**; both lines carry marks.

Worked example ✍ there is no largest prime

Claim: the primes are unbounded. **Suppose** $p_1, \ldots, p_m$ is a list of **all** the primes and let $P = p_1 p_2 \ldots p_m + 1$. Every p_i divides $P - 1$, so no p_i divides P.

$P > 1$ therefore has a prime divisor that is not on the list — contradiction.

So there is a prime above any given prime, and hence **no largest prime**.

Exam trap — assume the FULL negation

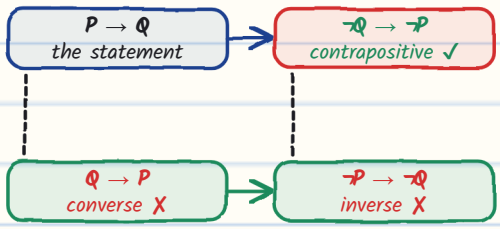
For a compound claim "every A is a B", the negation is **"there is an A that is not a B"** — not "not every A is a B" left hanging, and not "some A is not a C". Pushing on a partial negation is the commonest way a contradiction proof loses its last mark.

7 Contradiction vs Contrapositive

classic distinction ★

Definition To prove $P \to Q$ by the contrapositive you prove $\neg Q \to \neg P$. That is a different argument with the same power, because $P \to Q$ is logically equivalent to $\neg Q \to \neg P$. Contrapositive, by contrast, assumes $\neg(P \to Q) = P \wedge \neg Q$ — the hypothesis **true** and the conclusion **false** at once. The **converse** $Q \to P$ and the **inverse** $\neg P \to \neg Q$ are **not** equivalent to $P \to Q$, though they are equivalent to each other.

Form	Called	Equivalent to $P \to Q$?	Example with P = "n is even", Q = "n^2 is even"
$P \to Q$	the statement itself	—	true
$\neg Q \to \neg P$	contrapositive	**yes — always**	"n^2 odd ⇒ n odd" — true
$Q \to P$	converse	**no**	"n^2 even ⇒ n even" — false, n = 3
$\neg P \to \neg Q$	inverse	**no** (but converse ≡ inverse)	"n odd ⇒ n^2 odd" — true, yet not equivalent



only the top-right form always carries the top-left form

Exam trap The two mistakes that cost the mark: (i) calling $\neg Q \to \neg P$ the "converse" — it is the contrapositive; (ii) believing that a true inverse makes $P \to Q$ true. If both P and Q happen to be true the whole family is true, but equivalence is a statement about **all** cases, not one.

IT link — the same machinery, next chapter

Contradiction settles an **undecidability**: the **halting problem** is proved undecidable by assuming a decider D exists and asking D about D itself — D must accept exactly the machines that do **not** halt on D, a contradiction, so no D exists. Ordinary **induction** settles a **theorem**: a tree with n vertices has exactly **n − 1** edges — proved on n, using the leaf lemma (a longest path ends in a leaf) and deleting it.

One method proves a thing exists; the other proves nothing can.

Worked example ✍ the contrapositive doing the work

Prove: if n^2 is divisible by 4 then n is even.

Contrapositive: suppose n^2 is **not** divisible by 4; then n is odd, $n = 2k+1$, so $n^2 = 4k^2 + 4k + 1$, which is 1 more than a multiple of 4 — **not** divisible by 4.

The impossible remainder 1 is the ⊥.

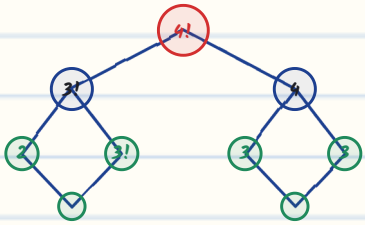
The four forms on one line $P \to Q$ the claim · $\neg Q \to \neg P$ the contrapositive, always equivalent · $Q \to P$ the converse, may fail · $\neg P \to \neg Q$ the inverse, may fail but equals the converse. Test: find **one** assignment where the two sides differ.

8 Recursion

definition ★

Definition A recursive definition defines an object **in terms of smaller instances of itself**; a recurrence relation does the same for sequences. Every recursive definition has exactly two parts: a set of **base cases** where the recursion stops, and a **recursive step** whose argument is **strictly closer to the base case**. A function that calls itself on an argument which does not decrease is not a recursion — it is an **infinite loop** that overflows the stack.

Object	Base case(s)	Recursive step
Factorial $n!$	$n! = 1$ if $n = 0$	$n! = n \times (n-1)!$ for $n > 0$
Fibonacci $F(n)$	$F(0) = 0$, $F(1) = 1$	$F(n) = F(n-1) + F(n-2)$ for $n > 1$
Power a^n	$a^0 = 1$	$a^n = a \times a^{n-1}$
Sum $S(n)$	$S(0) = 0$	$S(n) = S(n-1) + n$
Euclid $\gcd(a, b)$	$\gcd(a, 0) = a$	$\gcd(a, b) = \gcd(b, a \bmod b)$



the argument shrinks at every step

unrolling **factorial(4)**: every call moves to a smaller n and reaches n = 0

```
function factorial(n):
    if n == 0:              # base case — the recursion stops here
        return 1
    else:                   # step: n - 1 is strictly smaller than n
        return n * factorial(n - 1)
```

F(0)=0 F(1)=1 F(2)=1 F(3)=2 F(4)=3 F(5)=5 F(6)=8 F(7)=13 F(8)=21

Exam trap (i) **0! = 1** — the factorial base case is $n = 0$, not $n = 1$; since $n! = 1$ for both $n = 0$ and $n = 1$, checking only $n = 1$ does not verify the base case. (ii) Recursion and iteration compute the **same** result; neither is "the correct one", but naive recursion may be **exponentially slower** — see the next sheet. (iii) A **recurrence relation** (an $n \to n+1$ rule plus starting values) is **not** the same thing as a recursive algorithm; solving one is usually the other run backwards.

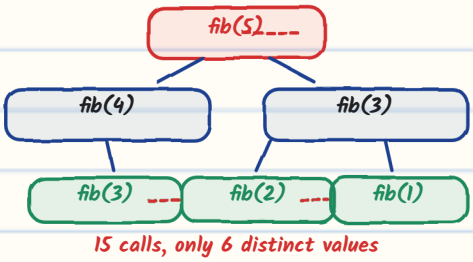
Fibonacci indexing The same sequence is written **F(0) = 0, F(1) = 1** or **F(1) = 1, F(2) = 1** — one index apart. Before answering an MCQ, check which convention the question uses; a question that says "F(1) = 1, F(2) = 1" is asking for the 7th value as **13**, not 8.

Why "smaller" is the whole condition Termination rests on a **well-founded measure**: some quantity — the argument n, a list size, a tree depth — must **strictly decrease** at every call and must have a **lower bound**. $\mathbb{N}$ has none, so repeated decrease forces arrival at 0 or 1. That is why the factorial step uses **n − 1** and not **n + 1**.

9 Recursion ↔ Induction, and What It Costs

high-yield ★

Why the two are one idea A recursive definition **supplies** the base cases and the step; a proof by induction **verifies** exactly those. So **to prove a recursive function correct, run induction on its argument**: base case — evaluate it at the base input; step — assume correctness for n, then the recursive call on **n−1** is covered by the hypothesis, and finish the algebra. This is the standard template for **algorithm correctness proofs**.



15 calls, only 6 distinct values

dashed = the same value computed twice

naive recursion re-computes the same sub-problems: exponential in n

Worked example ✍ the template, applied **Claim:** the function above returns n! for every $n \geq 0$. **Base:** factorial(0) returns 1 = 0!. **Step:** assume factorial(k) returns k! for some $k \geq 0$; then factorial(k+1) returns $(k+1) \times$ factorial(k) $= (k+1) \times k! = (k+1)!$. Hence by induction the function is correct for all $n \geq 0$. **Correctness proved; cost still open.**

Naive algorithm	Calls made	Cost
factorial(5)	5!, 4!, 3!, 2!, 1!, 0! = **6**	**O(n)** time, **O(n)** stack
fib(5), naive	$T(n) = T(n-1)+T(n-2)+1 \Rightarrow T(n) = 2F(n+1) - 1$; $T(5) = 15$	**O(φⁿ)**, $\varphi = (1+\sqrt{5})/2 \approx 1.618$
fib(5), memoised	each F(i) computed once: **n** calls	**O(n)** time

Exam trap "factorial is recursive so it must be slow" — **false**: it makes only n + 1 calls, i.e. **O(n)**, the same as a loop. The blow-up is **Fibonacci**, whose call tree branches into two. A **tail** call (the recursive call is the last operation) can be rewritten as a loop and then costs O(1) stack.

Where this earns marks These four lines — **correctness by induction**, **complexity of the call tree**, **program verification** and **theorem proving** — are the applications the syllabus names. Named systems built on them: **loop invariants** plus induction give total correctness of loops; **structural induction** on trees and lists proves compiler and datatype theorems; **diagonalisation** is contradiction applied to computability.

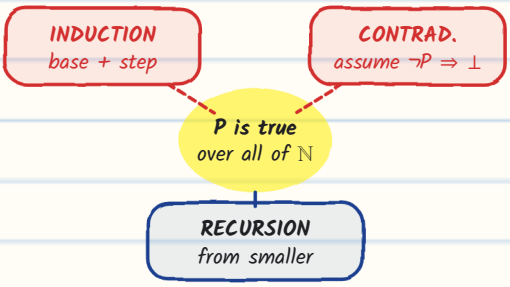
10 Which Method Do I Use?

the deciding table ★

Method	Key idea	You must supply	Fails when
Induction	prove true for a base case, then **step by step**	$P(b)$ true, and $P(k) \Rightarrow P(k+1)$ for **every** k	the claim is about $\mathbb{R}$, or the step needs several earlier cases
Contradiction	**assume false**, derive an impossible result	$\neg P$, then a genuine $\bot$ ($P \wedge \neg P$, $0 = 1$, $\div 0$)	you assume P itself, or derive something merely **unlikely**
Recursion	**define in terms of smaller instances**	base cases, plus a strictly decreasing step	the argument does not shrink — the recursion never terminates

The three in one line each **Induction** — "it is true once, and if it is true at k then it is true at $k+1$", so the truth spreads over all of $\mathbb{N}$. **Contradiction** — "if it were false we would get an impossibility", so it cannot be false. **Recursion** — "define it from the smaller ones", which is the **same information** the induction proof then verifies.

Induction and recursion are two halves of one object.



two proof techniques and one definition technique — they answer different questions

Applications — the four the syllabus names
Algorithm correctness proofs (induction on the loop counter or the recursion depth) · **Complexity analysis** (solve a recurrence by expansion or the master method) · **Program verification** (pre-condition, post-condition and invariants proved by induction) · **Mathematical theorem proving** (in a formal system a theorem is provable only if it follows from axioms by the inference rules of the previous lesson).

Exam trap The three are **not** three varieties of one thing: induction and contradiction are **proof techniques** — they argue about truth — while recursion is a **definition and a computational technique**. "Proving by recursion" is only meaningful as "proving the recursive program correct", and that proof is an **induction**. A claim with an easy **counterexample** is never proved by contradiction; it is simply **disproved**.

Night-Before-the-Exam Sheet

Fact sheet

Proof = **valid** + true premises = **sound**
Induction: **base case** $P(1)$
Step must hold for **every** $k \geq 1$
$\Sigma\ 1...n = \mathbf{n(n+1)/2}$
Strong induction uses $\mathbf{P(1)...P(k)}$
Contradiction: assume $\neg P$
$\sqrt{2}$ irrational: p, q both even
Contrapositive $\neg Q \to \neg P \equiv P \to Q$
$n! = 1$ $(n=0)$, else $n \cdot (n-1)!$
Recursion must **strictly decrease**
naive fib $T(n) = \mathbf{2F(n+1) - 1}$
memoise fib $\Rightarrow \mathbf{O(n)}$
Tree: n vertices $\Rightarrow \mathbf{n - 1}$ edges (induction)

Sound $\Rightarrow$ valid, never the reverse
Induction: **step** $P(k) \Rightarrow P(k+1)$
$P(k+1) \Rightarrow P(k)$ is not it
$\Sigma\ 2^0...2^n = \mathbf{2^{n+1} - 1}$
Strong $\equiv$ ordinary induction
$\bot = P \wedge \neg P$, $0 = 1$, $\div\ 0$
Then $\gcd(p, q) \geq 2 \neq 1$
Converse $Q \to P$ and inverse are not
$F(0)=0$, $F(1)=1$, else $F(n-1)+F(n-2)$
factorial(5) = **6** calls, $O(n)$
$T(5) = \mathbf{15}$, $O(\varphi^n)$, $\varphi \approx 1.618$
tail call $\Rightarrow O(1)$ stack
Halting undecidable (contradiction)

➜ Top 12 one-liners for MCQs

1. Induction needs **both** base case and step
2. The step is for **every** k, not one k
3. Contradiction assumes the **negation** of the goal
4. Contrapositive is **equivalent**; converse is not
5. $\mathbf{0! = 1}$ is the base case
6. Recursion needs a **smaller** argument
7. Naive fib is **exponential**, factorial is **linear**
8. A counterexample **refutes**, never proves
9. Finishing line: "**by the principle of induction**"

Top traps checking the inductive step at one value of k · writing $\mathbf{P(k+1) \Rightarrow P(k)}$ instead · starting the base case at $n = 1$ when the claim says $n \geq 0$ · assuming P instead of $\neg P$ in a contradiction proof · calling the contrapositive the converse · treating recursion as a proof rather than a definition · claiming naive factorial is slow · a recursive definition with no base case · "it held for the first few n, so it always holds".

All tricks together "**B**ase **B**uilds, the **S**tep carries it **every** time" · "**A**stract the **p**ositive: assume the **n**egation and push it to the absurd" · "recursion must be **smaller**, or it is not recursion at all" · "the **contra**positive is the twin; the **con**verse is the liar" · "one counterexample ends the argument" · "fib **branches**, factorial **stretches**" · "correctness by induction, termination by shrinking".

PROVES IT EXISTS
induction: $P(1)$ and $P(k) \Rightarrow P(k+1)$
n vertices $\Rightarrow n - 1$ edges

➜

PROVES NOTHING CAN
contradiction: assume $\neg P$, reach $\bot$
halting problem is undecidable

choose the method from the shape of the claim

one sentence decides which of the two proof techniques the question wants

✓ Practice MCQs (exam pattern, 10 marks)

try first, then check

1. A proof by contradiction begins by assuming:
(a) the statement is true (b) the converse
(c) the base case P(1) (d) the negation

2. In induction on P(n), the step must show:
(a) for every $k \geq 1$, $P(k) \Rightarrow P(k+1)$ (b) P(1) is true
(c) P(k + 1) for one particular k (d) $P(k+1) \Rightarrow P(k)$

3. $a(1) = 1$, $a(n) = 2a(n-1) + 1$. What is a(5)?
(a) 31 (b) 15
(c) 32 (d) 63

4. Which is equivalent to $P \to Q$?
(a) $Q \to P$ (b) $\neg P \to \neg Q$
(c) $\neg Q \to \neg P$ (d) $P \wedge Q$

5. $1 + 2 + 2^2 + \ldots + 2^n$ at $n = 0$ equals:
(a) 0 (b) 1
(c) 2 (d) 3

6. 0! equals:
(a) undefined (b) 1
(c) 0 (d) 2

7. Calls made by naive fib(5):
(a) 5 (b) 11
(c) 15 (d) 25

8. Calls made by naive factorial(5):
(a) 5 (b) 6
(c) 25 (d) 120

9. A single n where a claimed identity fails is:
(a) an induction step (b) a contradiction
(c) a counterexample (d) a base case

10. Always required of a recursive definition:
(a) it must use a loop (b) it calls itself on a smaller argument
(c) it must run in O(n) time (d) it must have a closed form

Answer key → 1-d · 2-a · 3-a · 4-c · 5-b · 6-b · 7-c · 8-b · 9-c · 10-b

Written question (i) State the principle of mathematical induction and prove $1 + 2 + 2^2 + \ldots + 2^n = 2^{n+1} - 1$ for every $n \geq 0$. (ii) Prove by contradiction that $\sqrt{2}$ is irrational. (iii) Give the recursive definition of the factorial function and the number of calls naive factorial(5) makes. (iv) Is the contrapositive of "n^2 even $\Rightarrow$ n even" true? 10

1. **PMI** — P(1) true + $P(k) \Rightarrow P(k+1)$ for every $k \geq 1 \Rightarrow P(n)$ for all $n \geq 1$. **Base n = 0:** $2^0 = 1 = 2^1 - 1$ ✓. **Step:** from $1 + \ldots + 2^k = 2^{k+1} - 1$, adding 2^{k+1} gives $2^{k+2} - 1$ ✓. 3
2. **√2** — suppose $\sqrt{2} = p/q$ in lowest terms; $p^2 = 2q^2 \Rightarrow p = 2k \Rightarrow q^2 = 2k^2 \Rightarrow q$ even. ~~Both even $\Rightarrow \gcd(p, q) \geq 2 \neq 1$ — contradiction.~~ ■ 3
3. **Factorial** — $n! = 1$ if $n = 0$, else $n \cdot (n-1)!$. Naive factorial(5) = **6** calls. 2
4. **Contrapositive** — "n^2 odd $\Rightarrow$ n odd" is **true** and equivalent; the original is **false** ($n = 3$). 2

Before the exam ✓ write **both** halves of an induction proof · say "**for every k ≥ 1**" in the step · start a contradiction with the **negation** · name the principle in the closing line.